

Lecture 15 Sampling Distribution And The Central Limit Theorem

BIO210 Biostatistics

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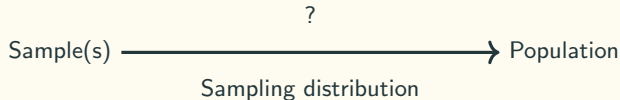
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Use Sample Statistics To Estimate Population Parameters



The scenario: we draw a sample of size n from the population. We observe the sample mean is $\bar{x}$ and the sample variance is s^2 . **We want to answer the following type of questions:**

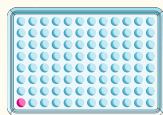
- **If the population mean were μ_0 :**

- what would be the probability of observing a sample of size n with a mean of $\bar{x}$?
- what would be the probability of observing a sample of size n with a mean falling into $[a, b]$?

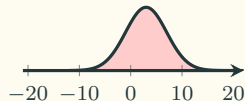
- **If the population variance were σ_0^2 :**

- what would be the probability of observing a sample of size n with a variance of s^2 ?
- what would be the probability of observing a sample of size n with a variance of $[a, b]$?

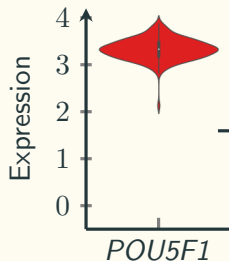
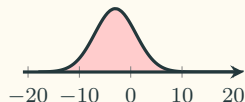
Intuition of Sampling Distribution



mESC: $\mathcal{N}(3, 4^2)$



Tc: $\mathcal{N}(-3, 4^2)$



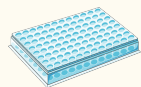
→ Which type?

Experiment_name

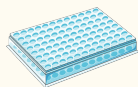
```
graph LR
    plate_1[plate_1] --- cell_1[cell_1]
    plate_1 --- cell_2[cell_2]
    plate_1 --- cell_3[cell_3]
    plate_1 --- dots1[...]
    plate_2[plate_2] --- cell_4[cell_1]
    plate_2 --- cell_5[cell_2]
    plate_2 --- cell_6[cell_3]
    plate_2 --- dots2[...]
```

Intuition of Sampling Distribution

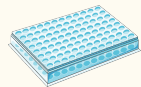
100 plates (samples)



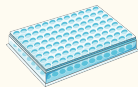
$n = 96$
 $\bar{x} = 3.07$



$n = 96$
 $\bar{x} = 2.97$

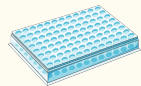


$n = 96$
 $\bar{x} = 2.80$



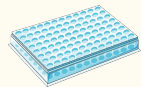
$n = 96$
 $\bar{x} = 4.95$

$\vdots$

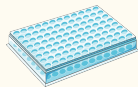


$n = 96$
 $\bar{x} = 3.60$

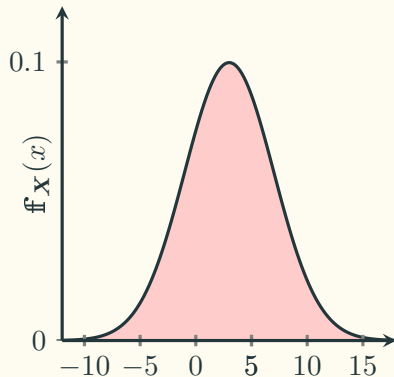
$\vdots$



$n = 96$
 $\bar{x} = 1.05$

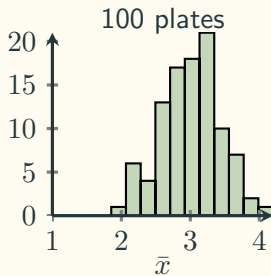
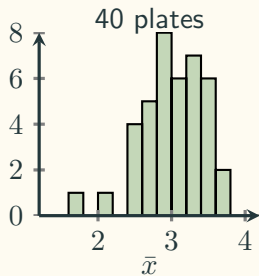
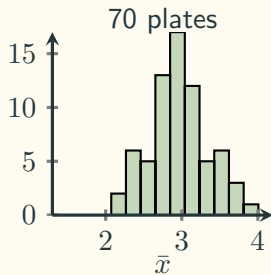
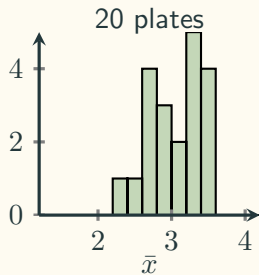
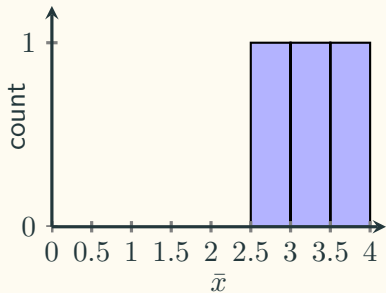
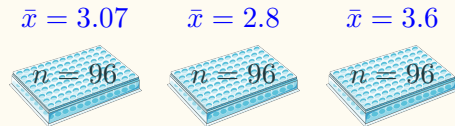


$n = 96$
 $\bar{x} = 3.27$

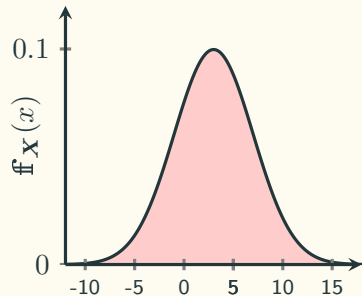


$$POU5F1 \sim \mathcal{N}(\mu = 3, \sigma^2 = 4^2)$$

Intuition of Sampling Distribution

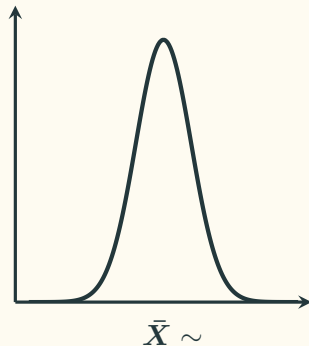


Sampling Distribution of The Sample Mean



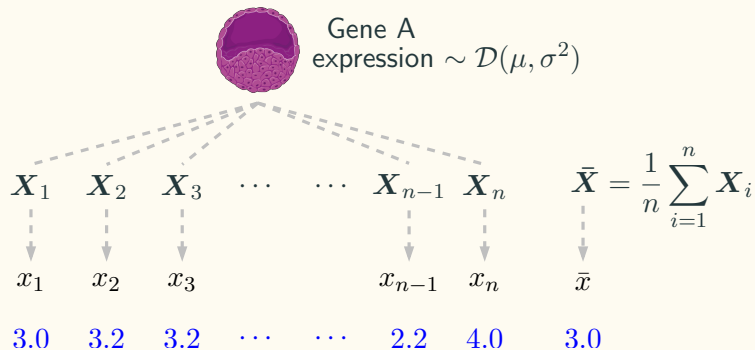
$$POU5F1 \sim \mathcal{N}(\mu = 3, \sigma^2 = 4^2)$$

$n = 96$
3.0



**Sampling distribution
of the sample mean**

i.i.d. Random Variables



$X_1, X_2, \dots, X_n$ are independent and identically distributed (**i.i.d.**) random variables.

$$X_1 \sim \mathcal{D}(\mu, \sigma^2)$$

$$X_2 \sim \mathcal{D}(\mu, \sigma^2)$$

$$X_3 \sim \mathcal{D}(\mu, \sigma^2)$$

$$\vdots$$

$$X_{n-1} \sim \mathcal{D}(\mu, \sigma^2)$$

$$X_n \sim \mathcal{D}(\mu, \sigma^2)$$

$$\bar{X} \sim ?(?, ?)$$

The Central Limit Theorem

By Pierre Simon de Laplace in 1810.

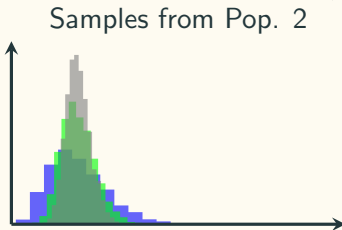
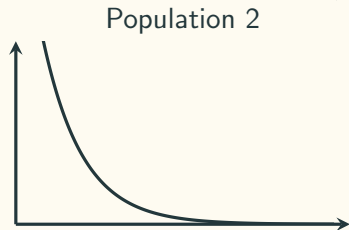
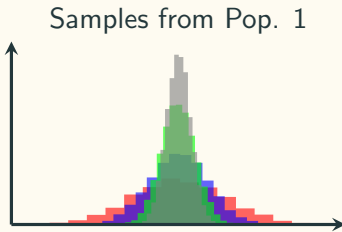
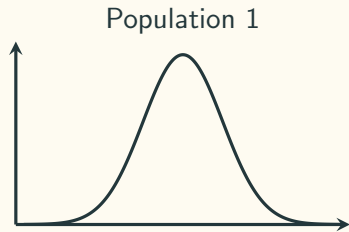
Theorem

The sampling distribution of the sample mean of n independent and identically distributed (i.i.d.) random variables is approximately normal, even if original variables themselves are not normally distributed, provided that n is large enough.

$$\bar{X} \sim \mathcal{N}(\mu_{\bar{X}}, \sigma_{\bar{X}}^2), \text{ where } \mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}: \text{ standard error.}$$

The Central Limit Theorem



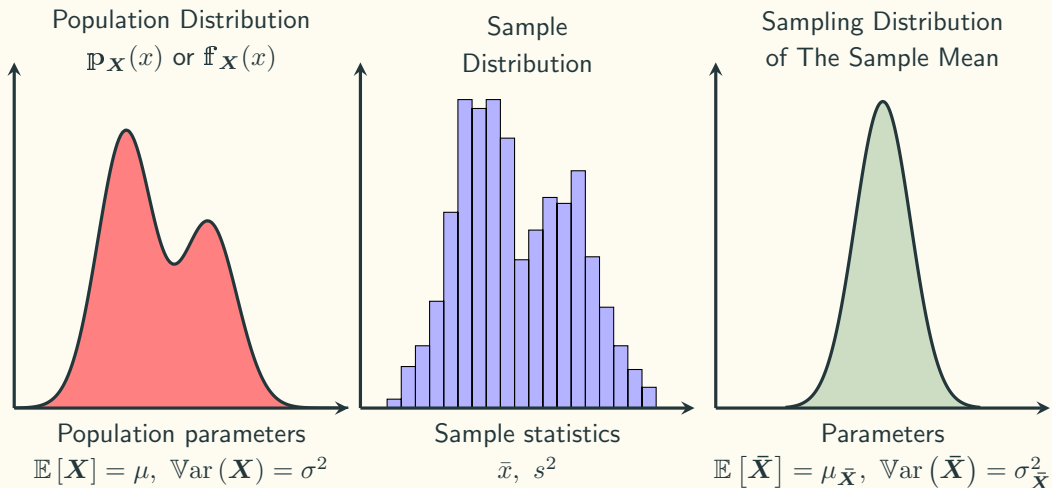
$n = 2$

$n = 5$

$n = 15$

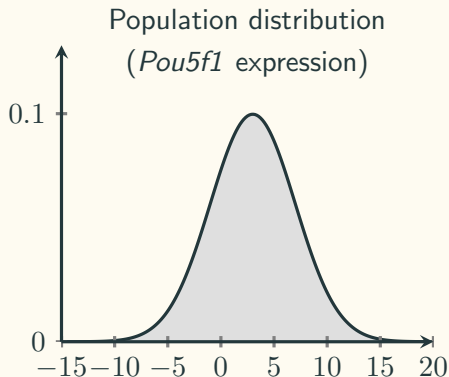
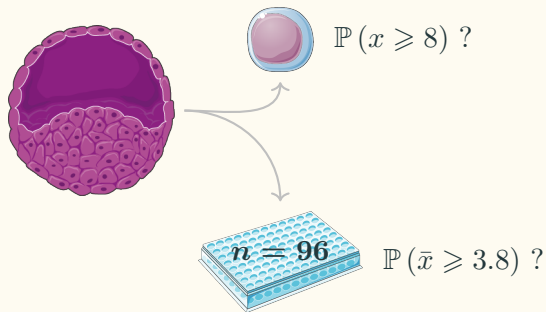
$n = 30$

Three Distributions

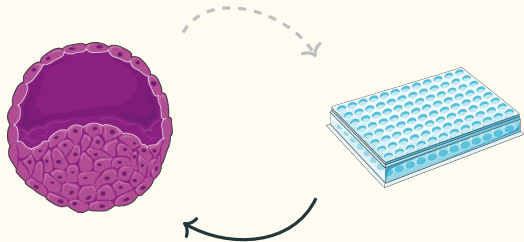


Practice: *Pou5f1* Expression

Based on the previous research, the expression of *Pou5f1* in all ES cells follow a normal distribution with $\mu = 3$ and $\sigma^2 = 4^2$.



Estimation



Use info. from the sample
to do a **point estimation**

Population parameter
 μ, σ^2

Sample statistics
 $\bar{x}, s^2$

- **Estimator**

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- **Estimate**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Unbiased Estimator

We say the following estimators are unbiased estimators:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Because:

$$\mathbb{E} [\bar{X}] = \mu$$

$$\mathbb{E} [S^2] = \sigma^2$$